

LayoutManager

```
import javax.swing.*;
import java.awt.*;

public class LayoutManagerDemo extends JFrame {

    private JPanel demoPanel;

    public LayoutManagerDemo() {
        setTitle("Swing Layout Manager Demo");
        setSize(600, 400);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        // Radio buttons
        JRadioButton flowButton = new JRadioButton("FlowLayout", true);
        JRadioButton borderButton = new JRadioButton("BorderLayout");
        JRadioButton gridButton = new JRadioButton("GridLayout");

        ButtonGroup group = new ButtonGroup();
        group.add(flowButton);
        group.add(borderButton);
        group.add(gridButton);

        JPanel radioPanel = new JPanel(new FlowLayout());
        radioPanel.add(flowButton);
        radioPanel.add(borderButton);
        radioPanel.add(gridButton);

        // Panel where the selected layout is demonstrated
        demoPanel = new JPanel();
        showFlowLayout();

        // Radio button actions
        flowButton.addActionListener(e -> showFlowLayout());
        borderButton.addActionListener(e -> showBorderLayout());
        gridButton.addActionListener(e -> showGridLayout());

        add(radioPanel, BorderLayout.NORTH);
        add(demoPanel, BorderLayout.CENTER);
    }

    private void showFlowLayout() {
        demoPanel.removeAll();
        demoPanel.setLayout(new FlowLayout(FlowLayout.CENTER, 15, 30));

        demoPanel.add(new JButton("Button 1"));
        demoPanel.add(new JButton("Button 2"));
        demoPanel.add(new JButton("Button 3"));
        demoPanel.add(new JButton("Button 4"));
        demoPanel.add(new JButton("Button 5"));
    }
}
```

```

        refreshPanel();
    }

    private void showBorderLayout() {
        demoPanel.removeAll();
        demoPanel.setLayout(new BorderLayout(10, 10));

        demoPanel.add(new JButton("NORTH"), BorderLayout.NORTH);
        demoPanel.add(new JButton("SOUTH"), BorderLayout.SOUTH);
        demoPanel.add(new JButton("EAST"), BorderLayout.EAST);
        demoPanel.add(new JButton("WEST"), BorderLayout.WEST);
        demoPanel.add(new JButton("CENTER"), BorderLayout.CENTER);

        refreshPanel();
    }

    private void showGridLayout() {
        demoPanel.removeAll();
        demoPanel.setLayout(new GridLayout(2, 3, 10, 10));

        demoPanel.add(new JButton("Button 1"));
        demoPanel.add(new JButton("Button 2"));
        demoPanel.add(new JButton("Button 3"));
        demoPanel.add(new JButton("Button 4"));
        demoPanel.add(new JButton("Button 5"));
        demoPanel.add(new JButton("Button 6"));

        refreshPanel();
    }

    private void refreshPanel() {
        demoPanel.revalidate();
        demoPanel.repaint();
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> {
            LayoutManagerDemo app = new LayoutManagerDemo();
            app.setVisible(true);
        });
    }
}

```

OUTPUT:

